Work, energy and power

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1. Topic Overview and Learning Objectives

Work, energy, and power form the fundamental pillars of classical mechanics, providing the language we use to describe how physical systems change over time. In our daily lives, we often use these terms interchangeably, but in physics, they have very specific, rigorous definitions that you must master to score well in your board exams. Understanding these concepts allows us to analyze everything from the motion of simple machines to the complex energy transformations occurring in our environment.

For the Sindh Board Class 9 Physics curriculum, these topics are high-yield areas that frequently appear in both objective and subjective sections. Examiners often test your conceptual clarity through numerical problems and short-answer questions. By grasping the relationship between force, displacement, and time, you will not only solve textbook problems but also develop a deeper intuition for how energy is conserved and transferred in the physical world around us.

The fundamental intuition here is that work is the process of energy transfer, energy is the capacity to perform that work, and power is simply the rate at which this transfer occurs. When you push an object, you are doing work on it, which changes its kinetic or potential energy. Mastering these definitions is the key to unlocking the rest of your physics studies, as these principles underpin thermodynamics, electricity, and even modern physics.

2. Core Concepts and Detailed Revision Notes

2.1 The Physics of Work

In physics, work is done only when a force applied to an object causes it to move in the direction of that force. This is a crucial distinction from our everyday language where we might feel tired after studying, but in physics, if there is no displacement, there is no work done. Mathematically, work is the product of the force applied and the displacement covered in the direction of the force. If the force is applied at an angle to the direction of motion, we must consider the component of the force that acts along the line of displacement. It is important to remember that work is a scalar quantity, meaning it has magnitude but no direction, even though force and displacement are vectors. Understanding this concept is the first step toward solving any mechanics problem involving energy changes.

Key Points to Remember:

1. Work is defined as the product of force and displacement in the direction of force.
2. The SI unit of work is the Joule, where 1 Joule equals 1 Newton-meter.
3. Work is a scalar quantity.
4. No work is done if the displacement is zero, regardless of the force applied.
5. Work done is zero if the force is applied perpendicular to the direction of motion.

2.2 Forms of Energy

Energy is defined as the ability of a body to do work. In our universe, energy exists in various forms, including mechanical, thermal, chemical, electrical, and nuclear energy. For your syllabus, we focus primarily on mechanical energy, which is divided into kinetic energy and potential energy. Kinetic energy is the energy possessed by an object due to its motion, depending on both its mass and the square of its velocity. Potential energy, on the other hand, is stored energy due to an object's position or configuration, such as gravitational potential energy stored in an object raised to a certain height. The Law of Conservation of Energy states that energy can neither be created nor destroyed, only transformed from one form to another. This principle is the backbone of all energy calculations in physics.

Key Points to Remember:

1. Kinetic energy is related to the motion of an object.
2. Gravitational potential energy is related to the height of an object above a reference point.
3. The total energy of an isolated system remains constant.
4. Energy is measured in Joules.
5. Mechanical energy is the sum of kinetic and potential energy.

2.3 Understanding Power

Power is defined as the rate at which work is done. While work tells us how much energy was transferred, power tells us how quickly that transfer happened. For example, two students might climb the same flight of stairs, doing the same amount of work, but the student who reaches the top first has exerted more power. Power is calculated by dividing the work done by the time taken to complete that work. The SI unit for power is the Watt, named after James Watt, and one Watt is equivalent to one Joule per second. In industrial applications, we often use larger units like the kilowatt or horsepower. Understanding power is essential for analyzing engines, motors, and human performance, as it highlights the importance of time in physical processes.

Key Points to Remember:

1. Power is the time rate of doing work.
2. The SI unit of power is the Watt.

3. 1 Watt is equal to 1 Joule per second.
4. Power is a scalar quantity.
5. Horsepower is a common unit used to measure the power of engines.

2.4 Efficiency and Energy Sources

Efficiency is a measure of how well a machine or system converts input energy into useful output energy. No machine is perfectly efficient because some energy is always lost to the surroundings, usually as heat due to friction or sound. Efficiency is expressed as a percentage, calculated by dividing the useful output energy by the total input energy and multiplying by 100. In the context of energy sources, we distinguish between renewable sources, like solar and wind, and non-renewable sources, like fossil fuels. As students of physics, it is important to understand that while we cannot create energy, we can improve the efficiency of our devices to reduce waste. This concept is vital for sustainable development and understanding the environmental impact of our energy consumption habits in the modern world.

Key Points to Remember:

1. Efficiency is the ratio of useful output to total input.
2. Efficiency is always less than 100 percent in real-world systems.
3. Energy loss is typically due to friction, heat, or sound.
4. Renewable energy sources are sustainable and environmentally friendly.
5. Improving efficiency helps in energy conservation.

3. Key Definitions, Formulas, and Rules

1. **Work:** W = F - d (Work equals force times displacement)
2. **Kinetic Energy:** KE = 0.5 - m - v^2 (Kinetic energy equals half mass times velocity squared)
3. **Potential Energy:** PE = m - g - h (Potential energy equals mass times gravity times height)
4. **Power:** P = W / t (Power equals work divided by time)
5. **Efficiency:** Efficiency = (Useful Output / Total Input) - 100
6. **Joule:** The work done when a force of 1 Newton moves an object through 1 meter.
7. **Watt:** The power used when 1 Joule of work is done in 1 second.
8. **Law of Conservation of Energy:** Energy cannot be created or destroyed, only transformed.
9. **Horsepower:** 1 hp = 746 Watts

4. Step-by-Step Worked Examples

Example 1: Calculating Work Done

Problem: A force of 50 N is applied to a box, moving it 10 meters in the direction of the force. Calculate the work done.

Solution Steps:

Step 1: Identify given values: Force F = 50 N, Displacement d = 10 m.

Step 2: Use the formula W = F - d.

Step 3: Substitute the values: W = 50 - 10.

Step 4: Calculate the result: W = 500 Joules.

Final Answer: 500 Joules

Example 2: Calculating Kinetic Energy

Problem: A car of mass 1000 kg is moving with a velocity of 20 m/s. Find its kinetic energy.

Solution Steps:

Step 1: Identify given values: Mass m = 1000 kg, Velocity v = 20 m/s.

Step 2: Use the formula KE = 0.5 - m - v^2.

Step 3: Substitute the values: KE = 0.5 - 1000 - (20^2).

Step 4: Calculate: KE = 500 - 400 = 200,000 Joules.

Final Answer: 200,000 Joules

Example 3: Calculating Potential Energy

Problem: An object of mass 5 kg is placed at a height of 10 meters. Calculate its potential energy (take g = 10 m/s^2).

Solution Steps:

Step 1: Identify given values: Mass m = 5 kg, Height h = 10 m, Gravity g = 10 m/s^2.

Step 2: Use the formula PE = m - g - h.

Step 3: Substitute the values: PE = 5 - 10 - 10.

Step 4: Calculate: PE = 500 Joules.

Final Answer: 500 Joules

Example 4: Calculating Power

Problem: A machine does 2000 Joules of work in 40 seconds. What is its power output?

Solution Steps:

Step 1: Identify given values: Work W = 2000 J, Time t = 40 s.

Step 2: Use the formula P = W / t.

Step 3: Substitute the values: P = 2000 / 40.

Step 4: Calculate: P = 50 Watts.

Final Answer: 50 Watts

5. Common Exam Mistakes to Avoid

1. Confusing work with effort: Remember that in physics, if there is no displacement, work is zero even if you are exhausted.
2. Forgetting units: Always write the units (Joules, Watts, kg, m/s) in your final answer to avoid losing marks.
3. Incorrectly squaring velocity: In the kinetic energy formula, ensure you square only the velocity, not the mass.
4. Ignoring the angle: If a force is applied at an angle, remember that only the component in the direction of motion does work.
5. Mixing up mass and weight: Potential energy uses mass (kg), not weight (Newtons). If weight is given, divide by g to get mass.

6. Practice Questions and Model Answers

Question 1

Define work and state its SI unit.

Answer 1: Work is the product of the force applied to an object and the displacement of the object in the direction of the force. Its SI unit is the Joule.

Question 2

Why is no work done when a person carries a heavy bag while walking on a level floor?

Answer 2: Work is done only if the force is in the direction of displacement. Here, the force (gravity) acts vertically downward, while the displacement is horizontal. Since the force is perpendicular to the displacement, the work done is zero.

Question 3

If the velocity of a body is doubled, what happens to its kinetic energy?

Answer 3: Since KE = 0.5 - m - v^2, if v becomes 2v, the new KE becomes 0.5 - m - (2v)^2 = 0.5 - m - 4v^2 = 4 - KE. The kinetic energy becomes four times the original value.

Question 4

A 100 W bulb is used for 10 seconds. How much energy is consumed?

Answer 4: Using P = W / t, we get W = P - t. W = 100 - 10 = 1000 Joules. The energy consumed is 1000 Joules.

Question 5

What is the primary cause of energy loss in machines?

Answer 5: The primary cause of energy loss in machines is friction, which converts useful mechanical energy into heat and sound energy.